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United States Patent	12397924
Kind Code	B2
Date of Patent	August 26, 2025
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Open rotor aircraft propulsion system with bypass flowpath

Abstract

A propulsion system for an aircraft includes an open propulsor rotor and a turbine engine configured to drive rotation of the open propulsor rotor about an axis. The turbine engine includes a core flowpath, a bypass flowpath, a fan section and an engine core. The engine core includes a compressor section, a combustor section and a turbine section. The core flowpath extends through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath. The inlet is next to and downstream of the fan section. The bypass flowpath extends outside of and along the engine core from an inlet into the bypass flowpath to an exhaust from the bypass flowpath. The inlet is next to and downstream of the fan section. The engine core is arranged axially between the fan section and the open propulsor rotor.

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Appl. No.: 18/431082

Filed: February 02, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250250020 A1	Aug. 07, 2025

Publication Classification

Int. Cl.: F02C3/06 (20060101); B64C11/48 (20060101); B64D27/10 (20060101); B64D33/08 (20060101); F02C6/20 (20060101)

U.S. Cl.:

CPC B64D33/08 (20130101); B64C11/48 (20130101); B64D27/10 (20130101); F02C3/06 (20130101); F02C6/206 (20130101);

Field of Classification Search

CPC: F02K (3/065); F02K (3/06); F02K (3/062); F02K (3/077)

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to an aircraft and, more particularly, to a propulsion system for the aircraft.

2. Background Information

(2) Various types and configurations of aircraft propulsion systems are known in the art including those with one or more open propulsor rotors. While these known aircraft propulsion systems have various benefits, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, a propulsion system is provided for an aircraft. This aircraft propulsion system includes an open propulsor rotor and a turbine engine configured to drive rotation of the open propulsor rotor about an axis. The turbine engine includes a core flowpath, a bypass flowpath, a fan section and an engine core. The engine core includes a compressor section, a combustor section and a turbine section. The core flowpath extends through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath. The inlet into the core flowpath is next to and downstream of the fan section. The bypass flowpath extends outside of and along the engine core from an inlet into the bypass flowpath to an exhaust from the bypass flowpath. The inlet into the bypass flowpath is next to and downstream of the fan section. The engine core is arranged axially between the fan section and the open propulsor rotor.

(4) According to another aspect of the present disclosure, another propulsion system is provided for an aircraft. This aircraft propulsion system includes an open propulsor rotor and a turbine engine configured to drive rotation of the open propulsor rotor about an axis. The turbine engine includes a core flowpath, a bypass flowpath, a fan section and an engine core. The engine core includes a compressor section, a combustor section and a turbine section. The core flowpath extends, in an axial direction towards the open propulsor rotor, through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath. The inlet into the core flowpath is downstream of the fan section. The bypass flowpath extends, in the axial direction towards the open propulsor rotor, from an inlet into the bypass flowpath to an exhaust from the bypass flowpath. The inlet into the bypass flowpath is downstream of the fan section.

(5) According to still another aspect of the present disclosure, another propulsion system is provided for an aircraft. This aircraft propulsion system includes an open propulsor rotor and a turbine engine configured to drive rotation of the open propulsor rotor about an axis. The turbine engine includes a core flowpath, a bypass flowpath, an exhaust flowpath, a fan section and an engine core. The engine core includes a compressor section, a combustor section and a turbine section. The core flowpath extends through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath. The inlet into the core flowpath is downstream of the fan section. The bypass flowpath extends from an inlet into the bypass flowpath to an exhaust from the bypass flowpath. The inlet into the bypass flowpath is downstream of the fan section. The exhaust flowpath fluidly couples the core flowpath and the bypass flowpath to an exhaust from the turbine engine.

(6) The bypass flowpath may extend from the inlet into the bypass flowpath to the exhaust from the bypass flowpath in a direction axially towards the open propulsor rotor.

(7) The inlet into the bypass flowpath may be radially outboard of the inlet into the core flowpath.

(8) The inlet into the core flowpath and the inlet into the bypass flowpath may each be configured as an annular inlet.

(9) The bypass flowpath may include an upstream section and a downstream section. The upstream section may be annular. The downstream section may be non-annular.

(10) The aircraft propulsion system may include a heat exchanger disposed in the downstream section.

(11) The downstream section may be a first downstream section, and the bypass flowpath may also include a second downstream section. The second downstream section may be non-annular. The first downstream section and the second downstream section may be fluidly coupled in parallel to the upstream section.

- (12) The aircraft propulsion system may include a first heat exchanger and a second heat exchanger. The first heat exchanger may be arranged with the first downstream section. The second heat exchanger may be arranged with the second downstream section.
- (13) The turbine engine may also include an inlet flowpath fluidly coupling an inlet into the turbine engine to the core flowpath and the bypass flowpath. The inlet flowpath may axially reverse direction between the inlet into the turbine engine and the fan section.
- (14) The turbine engine may also include an exhaust flowpath fluidly coupling the core flowpath and the bypass flowpath to an exhaust from the turbine engine. The exhaust flowpath may axially reverse direction between the engine core and the exhaust from the turbine engine.
- (15) The turbine section may include a free power turbine rotor. The free power turbine rotor may be coupled to and configured to drive the rotation of the open propulsor rotor.
- (16) The aircraft propulsion system may also include a geartrain coupling the free power turbine rotor to the open propulsor rotor.
- (17) The free power turbine rotor may be rotatable about the axis.
- (18) The axis may be a first axis, and the free power turbine rotor may be rotatable about a second axis that is offset from the first axis.
- (19) The engine core may include a rotating assembly rotatable about the axis. The rotating assembly may include a compressor rotor, a turbine rotor and a shaft coupling the turbine rotor to the compressor rotor. The compressor rotor may be in the compressor section. The turbine rotor may be in the turbine section.
- (20) The axis may be a first axis. The engine core may include a rotating assembly rotatable about a second axis offset from the first axis. The rotating assembly may include a compressor rotor, a turbine rotor and a shaft coupling the turbine rotor to the compressor rotor. The compressor rotor may be in the compressor section. The turbine rotor may be in the turbine section.
- (21) The aircraft propulsion system may also include an open guide vane structure downstream of the open propulsor rotor. The open guide vane structure may be configured to condition air propelled by the open propulsor rotor.
- (22) The aircraft propulsion system may also include a second open propulsor rotor. The turbine engine may be configured to drive rotation of the second open propulsor rotor about the axis.
- (23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
- (24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a sectional schematic illustration of a propulsion system for an aircraft.
- (2) FIG. 2 is a cross-sectional illustration of an open propulsor blade.
- (3) FIG. 3 is a cross-sectional illustration of an open guide vane.
- (4) FIG. 4 is a schematic plan view illustration of a bypass flowpath arranged with heat exchangers.
- (5) FIG. 5 is a sectional schematic illustration of the propulsion system with an angularly offset turbine engine.
- (6) FIG. 6 is a sectional schematic illustration of the propulsion system with a radially offset turbine engine.
- (7) FIG. 7 is a sectional schematic illustration of a portion of the propulsion system with a single rotor propulsion module.
- (8) FIG. 8 is a sectional schematic illustration of a portion of the propulsion system with a multi-rotor propulsion module.

DETAILED DESCRIPTION

- (9) FIG. 1 illustrates a propulsion system **20** for an aircraft. The aircraft may be an airplane, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The aircraft propulsion system **20** is configured as an open rotor propulsion system. The aircraft propulsion system **20** of FIG. 1, for example, extends axially along an axis **22** between an upstream, forward end **24** of the aircraft propulsion system **20** and a downstream, aft end **26** of the aircraft propulsion system **20**. The axis **22** may be a centerline axis of the aircraft propulsion system **20** and/or one or more of its members. The axis **22** may also or alternatively be a rotational axis of one or more members of the aircraft propulsion system **20**. The aircraft propulsion system **20** of FIG. 1 includes a propulsion module **28**, a turbine engine **30** and a heat exchange system **32**.
- (10) The propulsion module **28** of FIG. 1 includes an open propulsor rotor **34** and an open guide vane structure **36**. Here, the term “open” may describe a propulsion system section and/or a propulsion system component which is open to an environment **38** (e.g., an ambient environment) external to the aircraft propulsion system **20** and its propulsion module **28**. The propulsion module members **34** and **36** of FIG. 1, for example, are un-ducted and unshrouded components of the aircraft propulsion system **20** and its propulsion module **28**. The propulsion module **28** of FIG. 1 also includes a nose cone **40** disposed at (e.g., on, adjacent or proximate) the propulsion system forward end **24**. This nose cone **40** may be configured as a spinner which is rotatable with the propulsor rotor **34** about the axis **22**. Alternatively, the nose cone **40** may be configured as a stationary structure of the propulsion module **28**.
- (11) The propulsor rotor **34** includes a rotor base **42** (e.g., a disk or a hub) and a plurality of open propulsor blades **44** (e.g., airfoils). The propulsor blades **44** are arranged circumferentially around the rotor base **42** and the axis **22** in an array; e.g., a circular array. Each of the propulsor blades **44** is connected to (e.g., formed integral with or otherwise attached to) the rotor base **42**.
- (12) Each propulsor blade **44** projects spanwise along a span line of the respective propulsor blade **44** (e.g., radially relative to the axis **22**) out from an exterior surface **46** of the rotor base **42**, into the external environment **38**, to an unshrouded, distal tip **48** of the respective propulsor blade **44**. Each propulsor blade **44** is thereby configured as an un-ducted and unshrouded propulsor blade which is exposed to (e.g., disposed in) the surrounding external environment **38**. Referring to FIG. 2, each propulsor blade **44** extends longitudinally along a mean line **50** (e.g., a camber line) of the respective propulsor blade **44** from a leading edge **52** of the respective propulsor blade **44** to a trailing edge **54** of the respective propulsor blade **44**. Each propulsor blade **44** extends laterally between and to opposing side exterior surfaces **56A** and **56B** (generally referred to as “**56**”) of the respective propulsor blade **44**. The first blade exterior surface **56A** may be a concave, pressure side surface of the respective propulsor blade **44**. The second blade exterior surface **56B** may be a convex, suction side surface of the respective propulsor blade **44**. Each of these blade exterior surfaces **56** extends longitudinally along the blade mean line **50** between and meet at the respective blade leading edge **52** and the respective blade trailing edge **54**. Referring to FIG. 1, each blade element **52**, **54**, **56A**, **56B** extends spanwise from the base exterior surface **46** to the respective blade tip **48**.
- (13) Each propulsor blade **44** may be configured as a fixed propulsor blade. Each propulsor blade **44**, for example, may be fixedly connected to the rotor base **42**. Alternatively, some or all of the propulsor blades **44** may each be configured as a variable propulsor blade. Each propulsor blade **44**, for example, may be pivotally coupled to the rotor base **42** so as to be operable to change, for example, a pitch of the respective propulsor blade **44**.
- (14) The propulsor rotor **34** of FIG. 1 is arranged axially along the axis **22** between the nose cone **40** and the guide vane structure **36**. The propulsor rotor **34** of FIG. 1, in particular, is arranged axially downstream, aft of the nose cone **40** and axially upstream, forward of the guide vane structure **36**. An upstream, forward end of the propulsor rotor **34** may be disposed axially next to (e.g., adjacent, slightly spaced from) a downstream, aft end of the nose cone **40**. A downstream, aft

end of the propulsor rotor **34** may be disposed axially next to an upstream, forward end of the guide vane structure **36**.

(15) The guide vane structure **36** includes an inner platform **58** and a plurality of open guide vanes **60**; e.g., airfoils. The guide vanes **60** are arranged circumferentially around the inner platform **58** and the axis **22** in an array; e.g., a circular array. Each of the guide vanes **60** is connected to the inner platform **58**.

(16) Each guide vane **60** projects spanwise along a span line of the respective guide vane **60** (e.g., radially relative to the axis **22**) out from an exterior surface **62** of the inner platform **58**, into the external environment **38**, to an unshrouded, distal tip **64** of the respective guide vane **60**. Each guide vane **60** is thereby configured as an un-ducted and unshrouded guide vane which is exposed to (e.g., disposed in) the surrounding external environment **38**. Referring to FIG. **3**, each guide vane **60** extends longitudinally along a mean line **66** (e.g., a camber line) of the respective guide vane **60** from a leading edge **68** of the respective guide vane **60** to a trailing edge **70** of the respective guide vane **60**. Each guide vane **60** extends laterally between and to opposing side exterior surfaces **72A** and **72B** (generally referred to as “**72**”) of the respective guide vane **60**. The first vane exterior surface **72A** may be a convex, suction side surface of the respective guide vane **60**. The second vane exterior surface **72B** may be a concave, pressure side surface of the respective guide vane **60**. Each of these vane exterior surfaces **72** extends longitudinally along the vane mean line **66** between (and meet at) the respective vane leading edge **68** and the respective vane trailing edge **70**. Referring to FIG. **1**, each vane element **68**, **70**, **72A**, **72B** extends spanwise from the platform exterior surface **62** to the respective vane tip **64**.

(17) Each guide vane **60** may be configured as a fixed guide vane. Each guide vane **60**, for example, may be fixedly connected to the inner platform **58** and/or an internal support structure covered by the inner platform **58**. Alternatively, some or all of the guide vanes **60** may each be configured as a variable guide vane. Each guide vane **60**, for example, may be pivotally and/or otherwise moveably coupled to the inner platform **58** and/or the internal support structure covered by the inner platform **58** so as to be operable to change, for example, a pitch of the respective guide vane **60**.

(18) The guide vane structure **36** of FIG. **1** is arranged axially between the propulsor rotor **34** and one or more sections of the turbine engine **30**, or an entirety of the turbine engine **30**. The guide vane structure **36** of FIG. **1**, in particular, is arranged axially downstream, aft of the propulsor rotor **34** and axially forward of the turbine engine **30** and its engine sections. An aft end of the guide vane structure **36** may be disposed axially next to an upstream, forward end of the turbine engine **30**; e.g., relative to fluid flow within the external environment **38** outside of the aircraft propulsion system **20** and its turbine engine **30**.

(19) The turbine engine **30** of FIG. **1** includes an inlet section **74**, a fan section **75**, a compressor section **76**, a combustor section **77**, a turbine section **78** and an exhaust section **79**. This turbine engine **30** also includes an inlet flowpath **82**, a (e.g., annular) core flowpath **84**, a bypass flowpath **86** and an exhaust flowpath **88**.

(20) The compressor section **76** of FIG. **1** includes a low pressure compressor (LPC) section **76A** and a high pressure compressor (HPC) section **76B**. The turbine section **78** of FIG. **1** includes a high pressure turbine (HPT) section **78A**, an intermediate pressure turbine (IPT) section **78B** and a power turbine (PT) section **78C**, which PT section **78C** is a low pressure turbine (LPT) section of the aircraft propulsion system **20** and its turbine engine **30**. At least (or only) the LPC section **76A**, the HPC section **76B**, the combustor section **77**, the HPT section **78A**, the IPT section **78B** and the PT section **78C** may collectively form a core **90** of the turbine engine **30**; e.g., a gas generator.

(21) The engine sections **74**, **75**, **76A**, **76B**, **77**, **78A**, **78B**, **78C** and **79** may be arranged sequentially along the axis **22** between the propulsion system aft end **26** and the propulsion system forward end **24**. With this arrangement, each engine section **78A**, **78B**, **78C** and **79** is arranged axially along the axis **22** between (A) the propulsion module **28** and its components **34** and **36** and

(B) the combustor section **77**. The exhaust section **79** of FIG. **1** is also arranged axially along the axis **22** between (A) the propulsion module **28** and its components **34** and **36** and (B) the turbine section **78** and each of its sub-sections **78A-C**. The engine sections **75**, **76A**, **76B**, **77**, **78A**, **78B** and **78C** of FIG. **1** are arranged within an engine housing **92**; e.g., a case and a nacelle. The propulsor rotor **34** and the guide vane structure **36** are arranged outside of the engine housing **92**.

(22) Each of the engine sections **75**, **76A**, **76B**, **78A**, **78B** and **78C** includes a respective bladed rotor **94-99**; e.g., a ducted and/or shrouded engine rotor. Each of these engine rotors **94-99** includes a rotor base (e.g., a disk or a hub) and a plurality of rotor blades (e.g., airfoils, vanes, etc.). The rotor blades are arranged circumferentially around the respective rotor base and the axis **22** in an array. The rotor blades may also be arranged into one or more stages longitudinally along the core flowpath **84**. Each of the rotor blades is connected to the respective rotor base. Each of the rotor blades projects radially (e.g., spanwise) out from the respective rotor base into the core flowpath **84** and to a distal tip of the respective rotor blade.

(23) The HPC rotor **96** is coupled to and rotatable with the HPT rotor **97**. The HPC rotor **96** of FIG. **1**, for example, is connected to the HPT rotor **97** by a high speed shaft **102**. At least (or only) the HPC rotor **96**, the HPT rotor **97** and the high speed shaft **102** collectively form a high speed rotating assembly **104**; e.g., a high speed spool of the turbine engine **30** and its engine core **90**.

(24) The fan rotor **94** and the LPC rotor **95** are coupled to and rotatable with the IPT rotor **98**. The fan rotor **94** and the LPC rotor **95** of FIG. **1**, for example, are connected to the IPT rotor **98** by a low speed shaft **106**. At least (or only) the fan rotor **94**, the LPC rotor **95**, the IPT rotor **98** and the low speed shaft **106** collectively form a low speed rotating assembly **108**; e.g., a low speed spool of the turbine engine **30** and its engine core **90**.

(25) The PT rotor **99** (e.g., the LPT rotor) is connected to and rotatable with a power turbine shaft **110**. At least (or only) the PT rotor **99** and the power turbine shaft **110** collectively form a power turbine rotating assembly **112**.

(26) The propulsor rotor **34** is connected to and rotatable with a propulsor shaft **114**. At least (or only) the propulsor rotor **34** and the propulsor shaft **114** collectively form a propulsor rotating assembly **116**. This propulsor rotating assembly **116** is coupled to the power turbine rotating assembly **112** through a drivetrain **118**. The drivetrain **118** may be configured as a geared drivetrain, where a geartrain **120** (e.g., a transmission, a speed change device, an epicyclic geartrain, etc.) is disposed between and operatively couples the propulsor rotating assembly **116** and its propulsor shaft **114** to the power turbine rotating assembly **112** and its power turbine shaft **110**. With this arrangement, the propulsor rotor **34** may rotate at a different (e.g., slower) rotational velocity than the power turbine rotating assembly **112** and its PT rotor **99**. However, the drivetrain **118** may alternatively be configured as a direct drive drivetrain, where the geartrain **120** is omitted. With this arrangement, the propulsor rotor **34** and the PT rotor **99** may rotate at a common (the same) rotational velocity. Referring again to FIG. **1**, each of the rotating assemblies **104**, **108**, **112** and **116** and its members may be rotatable about the axis **22**.

(27) The inlet flowpath **82** fluidly couples an airflow inlet **122** into the turbine engine **30** from the exterior environment **38** to the core flowpath **84** and the bypass flowpath **86**. The inlet flowpath **82** of FIG. **1**, for example, extends longitudinally from the engine inlet **122**, through the fan section **75**, to an airflow inlet **124** into the core flowpath **84** and an airflow inlet **126** into the bypass flowpath **86**. Between the engine inlet **122** and the fan section **75**, the inlet flowpath **82** may axially reverse its direction. An upstream section of the inlet flowpath **82** of FIG. **1**, for example, projects axially in an axial aft direction (e.g., left-to-right in FIG. **1**) into the turbine engine **30** from the engine inlet **122**. The inlet flowpath **82** then turns radially inwards and reverses direction such that a downstream section of the inlet flowpath **82** extends axially in an axial forward direction (e.g., right-to-left in FIG. **1**) to the fan section **75**. This downstream section of the inlet flowpath **82** may also extend axially through the fan section **75** to the core inlet **124** and the bypass inlet **126**.

(28) The core flowpath **84** extends longitudinally, in the axial forward direction towards the

propulsion module **28** and its propulsor rotor **34**, through the engine core **90** from the inlet flowpath **82** to a core exhaust **128** from the core flowpath **84**. The core flowpath **84** of FIG. **1**, for example, extends sequentially through the LPC section **76A**, the HPC section **76B**, the combustor section **77**, the HPT section **78A**, the IPT section **78B** and the PT section **78C** from the core inlet **124** to the core exhaust **128**. The core inlet **124** of FIG. **1** is located downstream of and may be longitudinally/axially next to the fan section **75** and its fan rotor **94**. The core exhaust **128** of FIG. **1** is located downstream of and may be longitudinally/axially next to the PT section **78C** and its PT rotor **99**. The core inlet **124** and the core exhaust **128** may each be annular.

(29) The bypass flowpath **86** extends longitudinally, in the axial forward direction towards the propulsion module **28** and its propulsor rotor **34**, outside of the engine core **90** and its engine sections **76A-78C** from the bypass inlet **126** to one or more bypass exhausts **130** from the bypass flowpath **86**. The bypass flowpath **86** of FIG. **1**, for example, is disposed radially outboard of and extends axially along (e.g., axially overlaps) the engine core **90** from the bypass inlet **126** to the bypass exhausts **130**. The bypass inlet **126** of FIG. **1** is located downstream of and may be longitudinally/axially next to the fan section **75** and its fan rotor **94**. This bypass inlet **126** is disposed radially outboard of and may circumscribe the core inlet **124** where, for example, the bypass inlet **126** is annular. Each bypass exhaust **130** of FIG. **1** is located axially forward of the engine core **90** and its PT section **78C**. Each bypass exhaust **130** may be non-annular.

(30) Referring to FIG. **4**, the bypass flowpath **86** may include a flowpath feed **132** and one or more flowpath legs **134**. The flowpath feed **132** forms an upstream section of the bypass flowpath **86** which feeds (e.g., directs air into) the flowpath legs **134**. The flowpath feed **132** of FIG. **4**, for example, extends longitudinally from the bypass inlet **126** to each of the flowpath legs **134**.

Referring to FIG. **1**, the flowpath feed **132** may be annular and extend axially along and circumscribe the engine core **90** and its LPC section **76A**. The flowpath legs **134** form (e.g., parallel) downstream sections of the bypass flowpath **86** which laterally split off from the flowpath feed **132**; see also FIG. **4**. The flowpath legs **134** are disposed circumferentially about the engine core **90** and the axis **22**. Each flowpath leg **134** of FIG. **1**, for example, extends longitudinally from the flowpath feed **132** to its respective bypass exhaust **130**. Each flowpath leg **134** may be non-annular (e.g., arcuate, etc.) and extend axially along the engine core **90** and its engine sections **76B-78C**. It is contemplated, however, the bypass flowpath **86** may alternatively include a single annular downstream section which replaces the one or more (e.g., non-annular) flowpath legs **134**.

(31) The exhaust flowpath **88** fluidly couples the core flowpath **84** and the bypass flowpath **86** to an exhaust **136** from the turbine engine **30** into the exterior environment **38**. The exhaust flowpath **88** of FIG. **1**, for example, extends longitudinally from the engine core **90** at the core exhaust **128** to the engine exhaust **136**. The bypass exhausts **130** may be arranged longitudinally along the exhaust flowpath **88** at or (e.g., slightly) downstream of the core exhaust **128**. Between the engine core **90** and the engine exhaust **136**, the exhaust flowpath **88** may axially reverse its direction. An upstream section of the exhaust flowpath **88** of FIG. **1**, for example, extends axially in the axial forward direction away from the engine core **90**. The exhaust flowpath **88** then turns radially outwards and reverses direction such that a downstream section of the exhaust flowpath **88** extends axially in the axial aft direction to the engine exhaust **136**.

(32) With the foregoing arrangement, the core flowpath **84** and the bypass flowpath **86** are fluidly coupled in parallel between the inlet flowpath **82** and the exhaust flowpath **88**. The engine inlet **122** is located axially aft of the engine exhaust **136**. The engine inlet **122** and the engine exhaust **136** may be located to different (e.g., opposing) sides of the turbine engine **30**. In addition or alternatively, while the engine inlet **122** is shown in FIG. **1** as being axially aft of the engine exhaust **136**, it is contemplated additional ducting may be included for the engine inlet **122** and/or for the engine exhaust **136** to locate the engine inlet **122** axially forward of the engine exhaust **136**.

(33) During operation of the aircraft propulsion system **20**, ambient air within the external environment **38** is propelled by the propulsor rotor **34** across the guide vane structure **36** in an aft,

downstream direction towards the propulsion system aft end **26**. The guide vane structure **36** and its guide vanes **60** condition (e.g., straighten out) the air propelled by the propulsor rotor **34**, for example, to remove or reduce circumferential swirl. A major outer portion of the air propelled by the propulsor rotor **34** across the guide vane structure **36** provides forward thrust. A minor inner portion of the air propelled by the propulsor rotor **34** across the guide vane structure **36** is directed into the turbine engine **30** and its inlet flowpath **82** through engine inlet **122**. The air within the inlet flowpath **82** is further propelled (e.g., compressed) by the fan rotor **94** and directed into the core flowpath **84** and the bypass flowpath **86**. The air entering the core flowpath **84** may be referred to as “core air”. The air entering the bypass flowpath **86** may be referred to as “bypass air”.

(34) The core air is compressed by the LPC rotor **95** and the HPC rotor **96** and directed into a combustion chamber **138** (e.g., an annular combustion chamber) of a combustor (e.g., an annular combustor) in the combustor section **77**. Fuel is injected into the combustion chamber **138** and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially drive rotation of the HPT rotor **97**, the IPT rotor **98** and the PT rotor **99** before being exhausted from the engine core **90** into the exhaust flowpath **88**. The rotation of the HPT rotor **97** and the IPT rotor **98** respectively drive rotation of the HPC rotor **96** and the LPC rotor **95** and, thus, compression of the air received from the core inlet **124**. The rotation of the IPT rotor **98** also drives rotation of the fan rotor **94**. This rotation of the fan rotor **94** (A) propels the core air into the core flowpath **84** and (B) propels the bypass air through the bypass flowpath **86** into the exhaust flowpath **88**. The exhaust flowpath **88** subsequently directs the bypass air along with the combustion products out of the aircraft propulsion system **20** and its turbine engine **30** into the external environment **38** through the engine exhaust **136**. Concurrently, the rotation of the PT rotor **99** drives rotation of the propulsor rotor **34** through the drivetrain **118**. The rotation of the propulsor rotor **34** propels the ambient air within the external environment **38** across the guide vane structure **36** in the aft, downstream direction. With this arrangement, the turbine engine **30** and its engine core **90** power operation of (e.g., drive rotation of) the propulsor rotor **34** during aircraft propulsion system operation.

(35) The heat exchange system **32** of FIG. **1** includes one or more heat exchangers **140**. Each of these heat exchangers **140** may be arranged with (e.g., disposed in, project into, extend along, etc.) a respective one of the flowpath legs **134**; see also FIG. **4**. Each heat exchanger **140** is configured to transfer heat energy between the bypass air directed through the bypass flowpath **86** and a working fluid. Examples of the working fluid include, but are not limited to, the core air, the combustion products, lubricant, fuel, coolant, or the like. With the foregoing arrangement, the heat exchangers **140** utilize the bypass air rather than, for example, ambient air scooped from the external environment **38**, to enhance heat transfer efficiency. This may be particularly useful, for example, when the aircraft is operating on ground and air movement along the aircraft propulsion system **20** may be relatively slow and low pressure. By reducing or eliminating additional air scoop(s) along an exterior of the aircraft propulsion system **20**, free stream drag along the exterior of the aircraft propulsion system **20** may also be reduced. In addition, by directing the bypass air into the exhaust flowpath **88** following use with the heat exchange system **32**, remaining energy from the bypass air may be recuperated to supplement thrust generated by the combustion products exhausted through the engine exhaust **136** into the exterior environment **38**.

(36) In some embodiments, referring to FIG. **1**, the turbine engine **30** and its rotating assemblies **104**, **108** and **112** may be coaxial with the propulsion module **28** and its propulsor rotating assembly **116**. In other embodiments, referring to FIGS. **5** and **6**, one or more rotating assemblies **104**, **108** and/or **112** of the turbine engine **30** and its engine core **90** may rotate about an axis **142** which is offset from the rotational axis—the axis **22**—of the propulsor rotating assembly **116**. The engine axis **142** of FIG. **5**, for example, is angularly offset from the propulsor axis **22** by an included angle **144**. This angle **144** is a non-zero acute angle less than, for example, thirty or forty-five degrees (30-45°); e.g., between one degree (1°) and five degrees (5°), between five degrees

(5°) and ten degrees (10°), between ten degrees (10°) and twenty degrees (20°), etc. Here, the engine axis **142** is coincident with the propulsor axis **22**; however, the present disclosure is not limited to such an exemplary arrangement. In another example, referring to FIG. **6**, the engine axis **142** (and the power turbine axis) are laterally offset (e.g., spaced from) the propulsor axis **22**. The engine axis **142**, however, may (or may not) be parallel with the propulsor axis **22**.

(37) The aircraft propulsion system **20** may have various turbine engine configurations other than the ones described above. For example, while the turbine engine **30** is described above with two core rotating assemblies **104** and **108**, the turbine engine **30** may alternatively be configured with a single core rotating assembly (e.g., **104** or **108**) or three or more core rotating assemblies. In addition, while the PT rotor **99** is described above as a free turbine rotor, it is contemplated the PT rotor **99** may alternatively be coupled to one of the core rotating assemblies **104**, **108**, or another compressor rotor.

(38) The aircraft propulsion system **20** may have various propulsion module configurations other than the one described above. For example, referring to FIG. **7**, the propulsion module **28** may be configured with the single propulsor rotor **34** and without a guide vane structure. In another example, referring to FIG. **8**, the propulsion module **28** may be configured with multiple of the propulsor rotors **34A** and **34B** (generally referred to as “**34**”); e.g., counter-rotating open propulsor rotors. Each of these propulsor rotors **34** may be rotationally driven by the turbine engine **30** and its power turbine rotating assembly **112** (see FIG. **1**) through the drivetrain **118**. In the dual propulsor rotor arrangement of FIG. **8**, the propulsion module **28** is configured without a guide vane structure. The present disclosure therefore is not limited to the foregoing exemplary propulsor module configurations.

(39) The aircraft propulsion system **20** is described above with a tractor configuration; e.g., where the propulsor rotor **34** is disposed at or otherwise near the propulsion system forward end **24**. It is contemplated, however, the aircraft propulsion system **20** may be reversed to provide a pusher fan configuration. The present disclosure therefore is not limited to any particular open rotor propulsion system arrangement.

(40) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A propulsion system for an aircraft, comprising: an open propulsor rotor; and a turbine engine configured to drive rotation of the open propulsor rotor about an axis, the turbine engine including a core flowpath, a bypass flowpath, a fan section and an engine core, and the engine core including a compressor section, a combustor section and a turbine section; the core flowpath extending through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath, and the inlet into the core flowpath next to and downstream of the fan section; the bypass flowpath extending outside of and along the engine core from an inlet into the bypass flowpath to an exhaust from the bypass flowpath, and the inlet into the bypass flowpath next to and downstream of the fan section; and the engine core arranged axially between the fan section and the open propulsor rotor; wherein the bypass flowpath includes an upstream section and a downstream section, the upstream section is annular, and the downstream section is non-annular.

2. The propulsion system of claim 1, wherein the bypass flowpath extends from the inlet into the bypass flowpath to the exhaust from the bypass flowpath in a direction axially towards the open propulsor rotor.
3. The propulsion system of claim 1, wherein the inlet into the bypass flowpath is radially outboard of the inlet into the core flowpath.
4. The propulsion system of claim 1, wherein the inlet into the core flowpath and the inlet into the bypass flowpath each comprises an annular inlet.
5. The propulsion system of claim 1, further comprising a heat exchanger disposed in the downstream section.
6. The propulsion system of claim 1, wherein the downstream section is a first downstream section, and the bypass flowpath further includes a second downstream section; the second downstream section is non-annular; and the first downstream section and the second downstream section are fluidly coupled in parallel to the upstream section.
7. The propulsion system of claim 6, further comprising: a first heat exchanger arranged with the first downstream section; and a second heat exchanger arranged with the second downstream section.
8. A propulsion system for an aircraft, comprising: an open propulsor rotor; and a turbine engine configured to drive rotation of the open propulsor rotor about an axis, the turbine engine including a core flowpath, a bypass flowpath, a fan section and an engine core, and the engine core including a compressor section, a combustor section and a turbine section; the core flowpath extending through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath, and the inlet into the core flowpath next to and downstream of the fan section; the bypass flowpath extending outside of and along the engine core from an inlet into the bypass flowpath to an exhaust from the bypass flowpath, and the inlet into the bypass flowpath next to and downstream of the fan section; and the engine core arranged axially between the fan section and the open propulsor rotor, wherein at least one of the turbine engine further includes an inlet flowpath fluidly coupling an inlet into the turbine engine to the core flowpath and the bypass flowpath, and; and the inlet flowpath axially reverses direction between the inlet into the turbine engine and the fan section; or the turbine engine further includes an exhaust flowpath fluidly coupling the core flowpath and the bypass flowpath to an exhaust from the turbine engine, and the exhaust flowpath axially reverses direction between the engine core and the exhaust from the turbine engine.
9. The propulsion system of claim 1, wherein the turbine section comprises a free power turbine rotor; and the free power turbine rotor is coupled to and configured to drive the rotation of the open propulsor rotor.
10. The propulsion system of claim 9, further comprising a geartrain coupling the free power turbine rotor to the open propulsor rotor.
11. The propulsion system of claim 9, wherein the free power turbine rotor is rotatable about the axis.
12. The propulsion system of claim 9, wherein the axis is a first axis, and the free power turbine rotor is rotatable about a second axis that is offset from the first axis.
13. The propulsion system of claim 1, wherein the engine core comprises a rotating assembly rotatable about the axis; and the rotating assembly includes a compressor rotor, a turbine rotor and a shaft coupling the turbine rotor to the compressor rotor, the compressor rotor is in the compressor section, and the turbine rotor is in the turbine section.
14. The propulsion system of claim 1, wherein the axis is a first axis, and the engine core comprises a rotating assembly rotatable about a second axis offset from the first axis; and the rotating assembly includes a compressor rotor, a turbine rotor and a shaft coupling the turbine rotor to the compressor rotor, the compressor rotor is in the compressor section, and the turbine rotor is in the turbine section.

15. The propulsion system of claim 1, further comprising an open guide vane structure downstream of the open propulsor rotor, the open guide vane structure configured to condition air propelled by the open propulsor rotor.

16. The propulsion system of claim 1, further comprising: a second open propulsor rotor; the turbine engine configured to drive rotation of the second open propulsor rotor about the axis.

17. A propulsion system for an aircraft, comprising: an open propulsor rotor; a turbine engine configured to drive rotation of the open propulsor rotor about an axis, the turbine engine including a core flowpath, a bypass flowpath, an exhaust flowpath, a fan section and an engine core, and the engine core including a compressor section, a combustor section and a turbine section; the core flowpath extending through the compressor section, the combustor section and the turbine section from an inlet into the core flowpath to an exhaust from the core flowpath, and the inlet into the core flowpath downstream of the fan section; the bypass flowpath extending from an inlet into the bypass flowpath to an exhaust from the bypass flowpath, and the inlet into the bypass flowpath downstream of the fan section, wherein the inlet into the bypass flowpath is annular; and the exhaust flowpath fluidly coupling the core flowpath and the bypass flowpath to an exhaust from the turbine engine, wherein the inlet into the core flowpath is located axially aft of the exhaust flowpath.

18. The propulsion system of claim 17, wherein the inlet into the bypass flowpath is radially outboard of and circumscribes the inlet into the core flowpath.

19. The propulsion system of claim 17, wherein the fan section comprises a fan rotor radially overlapping the inlet into the bypass flowpath and the inlet into the core flowpath.
